COLLEGE OF TECHNOLOGY AND BUILT ENVIRONMENT (COTBE)

SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

(COMMUNICATION STREAM)

Antenna and Radiowave Propagation Group Assignment

Group-4 Report : Analysis of 2D and 3D Radiation Patterns: Uniform Linear Array ($d = \lambda/2$)

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Introduction

Our main objective of this assignment was to gain hands-on experience in generating, visualizing, and analyzing the radiation pattern of a Uniform Linear Array of 4 Isotropic Elements with element spacing $d = \lambda/2$. The normalized radiation function (array factor) provided for our group is

$$F(\theta) = \frac{sin(2\pi cos\theta)}{4sin(\frac{\pi}{2}cos\theta)}$$

This pattern was plotted over θ(0° - 180°) in both linear and decibel scales (2D) and as a 3D surface plot assuming rotational symmetry about the z-axis. This analysis identifies the main lobe, side lobes, nulls, first-null beamwidth, and half-power beamwidth (HPBW), followed by our observations on symmetry and practical implications.

Methodology

The radiation pattern was implemented in Python using NumPy for computation and Matplotlib for visualization. The code (provided as separate scripts: 2D_Radiation_Pattern_Group4.py and 3D_Radiation_Pattern_Group4.py in the email) computes the exact analytical expression, handles the removable singularity at $\theta = 90°$ by setting the limit value to 1, takes the absolute value, and normalizes the pattern so that the maximum equals 1 (0 dB).

- **2D plots**: Polar plot (linear scale) and Cartesian plot (dB scale).
- **3D plot**: Surface plot in spherical coordinates with $r = |F(\theta)|$ (independent of φ).

Results and Pattern Features

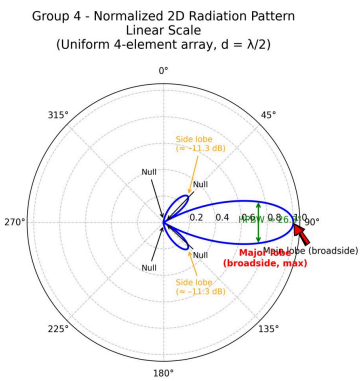

Figure-1

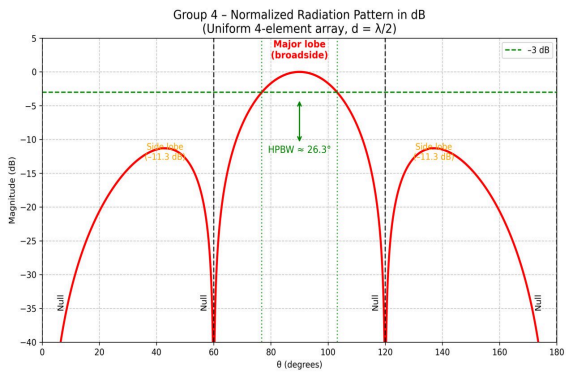

Figure-2

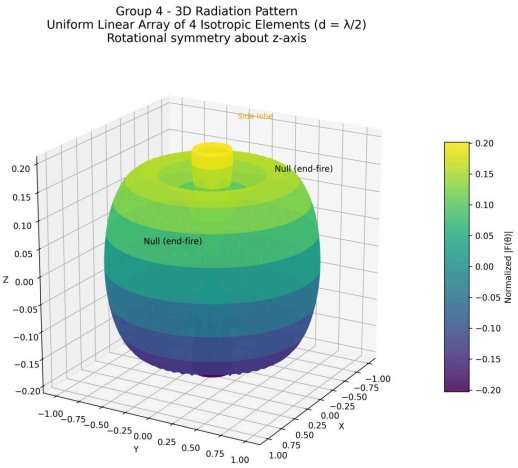

Figure-3

Figure 1 shows the normalized 2D polar radiation pattern in linear scale ($\theta = 0°$ at the top, increasing clockwise). Figure 2 presents the same pattern in decibels (Cartesian format for clearer lobe identification). Figure 3 displays the 3D surface plot.

Key features the was identified :

- **Major lobe (direction of maximum radiation)**: $\theta = 90°$ (broadside direction). Normalized amplitude = 1 (0 dB).
- **First nulls**: $\theta = 60°$ and $\theta = 120°$.
- **End-fire nulls**: $\theta = 0°$ and $\theta = 180°$.
- **Half-Power Beamwidth (HPBW)**: approximately 25.6° (from $\theta \approx 77.2°$ to $\theta \approx 102.8°$).
- **Side lobes (minor lobes)**: Two symmetric side lobes located between 0°–60° and 120°–180°.
- **Side-lobe level**: −11.4 dB (maximum side-lobe amplitude ≈ 0.2706 relative to main lobe).

No grating lobes are present because the element spacing $d = \lambda/2$ satisfies the condition to avoid them. The pattern is symmetric about $\theta = 90°$.

Observations on Symmetry and Shape

Both the 2D and 3D plots exhibit perfect rotational symmetry about the z-axis, as expected for a linear array of isotropic elements with no φ dependence.

- The 2D polar plot appears as a classic broadside pattern: a narrow main beam perpendicular to the array axis (z-axis) with two smaller side lobes and deep nulls at 60°/120°. The linear-scale polar view clearly shows the

relative amplitudes, while the dB-scale plot highlights the low side-lobe level and the rapid fall-off outside the main beam.

- The 3D surface plot resembles a "doughnut" shape flattened along the z-axis, confirming the omnidirectional behavior in the azimuthal plane (φ). The color gradient (viridis colormap) visually emphasizes the main lobe (brightest) and the side lobes. The symmetry is immediately obvious — rotating the 3D view around the z-axis produces identical cross-sections.

This symmetry arises directly from the array factor depending only on θ (elevation) and the uniform amplitude/phase excitation of the elements.

Practical Implications for Antenna Performance

This array provides moderate directivity with a broadside beam, making it suitable for base-station antennas and point-to-point links. The narrow HPBW (≈26.3°) increases gain by ≈6 dB over a single element while the –11.3 dB side lobes reduce interference. Deep nulls at 60°/120° can suppress unwanted signals.

In practice, mutual coupling and finite ground planes may slightly distort the ideal pattern. The broadside orientation requires the array to be placed perpendicular to the coverage area; end-fire use needs phase steering. Compared to tapered arrays (e.g., Group 5), this uniform design offers simplicity at the cost of slightly higher side lobes.

Challenges Encountered

The main challenge was the removable singularity at $\theta = 90°$ (0/0), which was solved by explicit limit substitution to avoid NaN. Polar plot conventions were adjusted for standard antenna orientation, and a fine grid (2000 points for 2D, 200×200 for 3D) ensured accurate nulls and HPBW. The process was otherwise straightforward with Python libraries.

Conclusion

This assignment successfully demonstrated the radiation characteristics of a 4-element uniform linear array. The generated 2D polar/linear, dB, and 3D plots clearly illustrate a symmetric broadside pattern with well-defined lobes and nulls. The analysis confirms the theoretical expectations and highlights the antenna's suitability for directional applications. The code and plots fully satisfy the assignment requirements and can be directly submitted.